Bipolaris-RNAi Designer (BRD) v3.1 — Final Validation Report

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Branch / Release: `v3.1` (`11_post_execution_audit_v3.1/`)

Scope Notice: Computational target evaluation, orthology, conservation, off-target screening, and heuristic prioritization only. No wet-lab silencing, exogenous dsRNA uptake in *Bipolaris sorokiniana*, crop protection, or environmental biosafety is claimed.

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1. Executive summary

BRD v3.1 upgrades the reproducible v3 pipeline across software audit, native Windows 11 execution (`Broccoli v2.0.1` , `Biopython 1.86` assembly QC), dataset verification (`pleosporales_odb12` , `fungi_odb12` , reference FASTA SHA-256s, exact 20/21 mismatch positions), dependency safety (guarded optional `ViennaRNA`), and clean-extraction verification (`hydrate_inputs_v3_1.py` + synchronized `hashes_v3_1.tsv`). Core biological conclusions and the 8-region clean candidate shortlist remain unchanged (`BRD_CsSTE11_CAND013` , `0.9418` , Pareto Tier 1), while `Broccoli v2.0.1` newly resolves true 1:1 orthology for `CsFUS3` (`OG_1315`) and `CsPKS1` (`OG_4251`) and empirical `compleasm run` limitations on native Windows are logged without synthetic scores.

2. v3 baseline findings

Stage 0 (`stage0_audit/V3_Baseline_Audit_Report.md`) preserved all original v1 (`10_final_reports/`), v2 (`11_post_execution_audit/`), and v3 (`11_post_execution_audit_v3/`) artifacts and confirmed all `10` independent audit issues (`AUDIT_01` – `AUDIT_10`), notably missing input genomes inside `11_post_execution_audit_v3.zip` (`AUDIT_05`), `hashes_v3.tsv` including 3 skipped Stage 2 files (`AUDIT_07`), and unexecuted Linux binaries (`AUDIT_01` – `04`).

3. v3.1 changes

See full auditable details in `V3_1_Change_Log.md` . Principal additions: `assembly_qc_biopython.py` , `stage4_compleasm_runner.py` , native compiled `tools/bin/broccoli.exe` (`v2.0.1`) + `stage5_broccoli_orthology.py` , `stage6_conservation_v31.py` , `stage7_offtarget_v31.py` , `stage8_to_11_v31.py` , `hydrate_inputs_v3_1.py` , `check_environment_v3_1.py` , Docker/Remote Linux templates, and `test_brd_v3_1.py` .

4. Tool investigation results

Repository	Version	License	Status
https://github.com/biopython/biopython	Installed 1.86 (upstream latest tag: biopython-188)	Biopython License Agreement / 3-Clause BSD (dual-licensed modules)	PASSED (Native Windows 11 Route A — all 19 isolates parsed & unit-tested; not QUAST)
https://github.com/huangnengCSU/compleasm	v0.2.10 (commit a275e9f4, released 2026-10-07)	Apache-2.0 (+ BUSCO MIT / CC-BY-ND-4.0 dataset license)	BLOCKED on native Windows analysis (CLI/list/download PASSED rc=0; run BLOCKED rc=1 missing Win32 miniprot/hmmsearch; Docker/WSL2 absent)
https://github.com/rderelle/Broccoli	v2.0.1 Rust rewrite (commit e54756aa, released 2026-10-01); v1.4.0 Python audited	GPL-3.0-or-later	PASSED (Native Windows 11 Route A with -phylogenies nj; default -phylogenies bionj FastTree mode BLOCKED due to upstream MicrobesOnline anti-bot gate)

Execution routes verified (`execution_environment/execution_routes_status.tsv`):

route	availability	blocked_tools
Route_A_Native_Windows_11	ACTIVE	compleasm run (missing Win32 miniprot/hmmsearch), Broccoli bionj (FastTree blocked), QUAST, BUSCO, OrthoFinder, RNAhybrid
Route_B_Docker_Desktop	UNAVAILABLE (docker CLI missing)	All container targets until Docker Desktop is installed
Route_C_Remote_Linux_SSH	UNCONFIGURED (no remote host/keys)	All remote Linux targets until SSH endpoint is supplied

5. Genome QC results (Biopython — Not QUAST)

All `19/19` *B. sorokiniana* assemblies parsed cleanly (`0` duplicate IDs, `0` duplicate contig sequences, `0` invalid nucleotide bases). Assemblies span `32.42–38.20 Mb` , N50 `130.08 kb–2.31 Mb` , L50 `7–73` . Two scaffolded assemblies (`ND90Pr` `3.4774%` N, `Sacc._Shoemaker` `3.4789%` N) triggered `ELEVATED_AMBIGUOUS_N_PCT` warnings (`assembly_qc_warnings_v3_1.tsv`). Intrinsic Biopython statistics **do not** detect misassemblies or local consensus homopolymers (`WAI2406` , `WAI2411`).

6. Compleasm results

lineage	orthodb_version	marker_count	verified_tar_md5	miniprot_version	hmmer_version
pleosporales_odb12	OrthoDB_v12.1	6066	795a5af234fb9b7828b9927dfbebff4a	MISSING (Linux x64 binary only; unavailable natively on Windows)	MISSING (HMMER 3.4 POSIX-only build; unavailable natively on Windows)
fungi_odb12	OrthoDB_v12.1	1122	79aba415d64ed9a7b1994452b99106cc	MISSING (Linux x64 binary only; unavailable natively on Windows)	MISSING (HMMER 3.4 POSIX-only build; unavailable natively on Windows)

- **Empirical result:** `compleasm.py v0.2.10` CLI (`-h` , `list --remote` , `download`) succeeded on native Windows 11, downloading and verifying `pleosporales_odb12` (`6,066` markers) and `fungi_odb12` (`1,122` markers). Executing `compleasm.py run` on `ND90Pr` exited `rc=1` due to missing native Windows `miniprot` and `hmmsearch`

(`compleasm_execution_log.txt`). All `19` rows in `compleasm_summary_v3_1.tsv` remain `NOT_RUN_DEPENDENCY_BLOCKED` (`NA` scores; never substituted or fabricated). Separate 50-marker tBLASTn core frame audit remains referenced in `compleasm_marker_details_v3_1.tsv` .

7. Broccoli results

- Compiled **Broccoli v2.0.1** (`GPL-3.0-or-later` , Rust `1.99.1` `x86_64-pc-windows-gnu`) natively to `tools/bin/broccoli.exe` and executed Steps 1–4 with `diamond.exe v2.2.8` using built-in `-phylogenies nj` (smoke run `4.07s` , full 36,714-protein 3-species run `255.98s` , `rc=0`).
- **Results:** `9,589` orthologous groups (`8,793` shared between `ND90Pr` and `Sacc._Shoemaker` ; `5,710` across all 3 proteomes; `26,745` orthologous pairs; `0` chimeric proteins flagged). *Broccoli orthogroups are strictly labelled Broccoli (`OG_`), never OrthoFinder.**

8. DIAMOND RBH comparison

metric	analysis_a_diamond_rbh_v3	analysis_b_broccoli_v2_nj
Total orthogroups inferred	15015	9589
Core orthogroups in BOTH B. sorokiniana proteomes (ND90Pr + Sacc.)	7245	8793
Three-species conserved orthogroups (+ Botrytis cinerea)	4430	5710
Proteins assigned to any group (unique IDs)	36714	28051
Broccoli orthologous pairs (step 4)	NA (connected components)	26745
Broccoli ortholog pairs co-clustered in same DIAMOND RBH OG	22915	22915/26745 (85.68%)
Chimeric proteins flagged by Broccoli	nan	0

target	nd90pr_protein_id	broccoli_og	diamond_rbh_og
CsPKS1	XP_007696213.1	OG_4251	RBH_OG000102
CsFUS3	XP_007697446.1	OG_1315	RBH_OG000002
CsSTE11	XP_007694574.1	OG_1786	RBH_OG002545
CsSp1	XP_007704210.1	OG_9092	RBH_OG008444

Whereas DIAMOND RBH graph clustering merged `CsFUS3` and `CsPKS1` into large kinase (`RBH_OG000002`) and PKS (`RBH_OG000102`) paralog superfamilies, Broccoli's tree/network partitioning isolated clean 1:1 orthologs (`OG_1315` , `OG_4251`). Concordance (`85.68%` = `22,915/26,745` pairs co-clustered) confirms broad agreement without constituting independent biological proof.

9. Conservation results

target	isolates_complete_cds	msa_min_nt_identity_vs_nd90pr_pct	msa_ambiguous_n_bases_total	candidate_level_min_nt_id_hq17_pct	candidate_level_min_21mer_hq17_pct
CsPKS1	17	99.8452	0	99.44	86.88
CsFUS3	17	99.9053	0	99.5	90.87
CsSTE11	17	99.5832	0	99	78.89
CsSp1	19	99.2593	0	98.89	76.67

Full 50x19 identity/21-mer matrices and lowest-identity isolate labels (`LK93` reducing `CsSp1_CAND008` to `79.38%` HQ-17 21-mer preservation; `WAI2406` / `WAI2411` homopolymer indels isolated) are reported in `stage6_conservation/` .

10. Off-target results

Verified local FASTA SHA-256 checksums across all 18 reference panel entries (`stage7_offtarget/offtarget_reference_metadata.tsv`), resolved strand/coordinates across `1,214` exact 21-mer hit rows (`18/50` excluded, `32/50` pass), and resolved full 21-nt mismatch positions across `16,073` near-hits (`20/21` : `3,655` ; `19/21` : `7,501` ; `18/21` : `4,917`). Wheat, barley, and maize nuclear non-coding genomes (`>20 GB`) and unsequenced field microbes remain unscreened. **Zero exact matches never establish biological safety, and `7/8` shortlist leaders carry elevated `20/21` near-match burdens (`6–14` guides).**

11. Candidate-region results

`49/50` candidates match contiguous authoritative target CDS coordinates/substrings on `BRD_Target_Gene_Sequences.fasta` cleanly (`1` candidate, `BRD_CsSTE11_CAND009` from `dsRNAmax` , is a stitched `SYNTHETIC_KMER_CONSTRUCT`), and all `50` map to `12` independent non-redundant target regions (`8` clean passing). Sequence-derived descriptors, `ViennaRNA 2.8.0` structural metrics, blocked `RNAhybrid` status, and heuristic targetability scores remain cleanly separated (`stage8_candidate_design/candidate_master_v3_1.tsv`).

12. Biological target evidence

gene_id	biological_relevance_score	off_target_concerns	confidence_level
CsSTE11	0.9921	Region 04 (2226..2525) fails exact 21-mer screening in beneficial fungi; Regions 01/03 low near-match signal (3-4 20/21 hits); Region 02 has 14 20/21 near-hits	HIGH (direct B. sorokiniana KO + PHI curation + 1:1 Broccoli orthology)
CsPKS1	0.5796	0 exact non-target 21-mers across 4 candidate regions, but ALL 4 regions carry Elevated_20of21_Near_Match_Burden (6–12 unique 20/21 guides)	MODERATE-HIGH for RNAi tractability, LOW for disease suppression (direct KO proves unaffected pathogenicity)
CsSp1	0.55	0 exact 21-mers across panel, 8 unique 20/21 near-matches; short 270-nt ORF leaves no alternate non-overlapping region	MODERATE (strong direct wheat-leaf KO & valid 3B/4B/5B induction, but missing PHI/InterPro & SNP drop in LK93)
CsFUS3	0.9921	SEVERE BENEFICIAL-FUNGI & CROP HOMOLOGY: 13/13 candidates excluded by >=1 exact 21-mer match across Trichoderma/Metarhizium/Beauveria/Rice/Wheat/Maize; exhaustive CDS sweep shows only 5/1059 windows escape exact hits	HIGH biological relevance / EXCLUDED for selective dsRNA design across existing windows

- `CsSTE11` leads (`0.9921`); `CsPKS1` preserves `PHI:2260` `unaffected pathogenicity` ; `CsSp1` retains `3B/4B/5B` expression (`max = 139.61 RPKM` , invalid `1,438.44` excluded); `CsFUS3` retains beneficial-fungi homology exclusion.

13. Ranking results

`independent_region_ranking_v3_1.tsv` orders regions by `(passes_zero_exact_21mer_filter DESC, pareto_front_3d_regions ASC, heuristic_composite_prioritization_index DESC)` whereas pure composite ranking ignores Pareto tier dominance — explaining why Tier-1 `BRD_CsPKS1_CAND010` (`0.7762`) places `#5` ahead of Tier-2 `BRD_CsPKS1_CAND013` (`0.7841` , `#6`).

14. Literature benchmarking

Audited `25` curated benchmark studies (`14` gene-level only, `5` ambiguous/conflicting, `4` computational only, `2` reconstructed *P. pachyrhizi* EST sequences) plus `8` exploratory *F. graminearum* constructs (`stage11_literature_benchmark/literature_benchmark_v3_1.tsv`). *B. sorokiniana* still lacks any published exogenous dsRNA SIGS uptake assay.

15. Test results

Automated suite summary recorded in `V3_1_Test_Results.txt` and `V3_1_Test_Summary.md` (covering both workspace regression tests and independent clean-extraction execution).

16. Reproducibility assessment

Clean extractions can verify all packaged SHA-256 hashes immediately (`hashes_v3_1.tsv` synchronized 1:1 with ZIP entries), inspect environment readiness (`check_environment_v3_1.py`), and hydrate missing NCBI genomes with verified SHA-256 checksums (`hydrate_inputs_v3_1.py --primary-only` or all 19).

17. Remaining limitations

1. 1. Linux-only external tools (`compleasm run` needing `miniprot` / `hmmsearch` , `QUAST` , `BUSCO` , `OrthoFinder` , `RNAhybrid` , `FastTree bionj`) remain blocked natively on Windows 11 without Docker or WSL2.
2. 2. Computational rankings do not demonstrate exogenous dsRNA uptake, in vivo knockdown, or disease reduction in *B. sorokiniana*.
3. 3. Whole nuclear non-coding genomes of wheat, barley, and maize remain outside local off-target coverage.
4. 4. `7/8` clean independent-region leaders retain elevated `20/21` near-match burdens (`6–14` guides) alongside `CAND013` (`3`) and `CAND014` (`4`).

18. Final computational candidate shortlist

independent_region_rank_v3_1	composite_index_rank_within_filter_tier	candidate_id	target	region_id	candidate_length_nt	biological_target_relevance_score	dsrna_targetabilit
1	1	BRD_CsSTE11_CAND013	CsSTE11	CsSTE11_REG01	210	0.9921	0.8855
2	2	BRD_CsSTE11_CAND014	CsSTE11	CsSTE11_REG03	210	0.9921	0.8859
3	3	BRD_CsSTE11_CAND010	CsSTE11	CsSTE11_REG02	180	0.9921	0.8913
4	4	BRD_CsPKS1_CAND012	CsPKS1	CsPKS1_REG02	210	0.5796	0.8879
5	7	BRD_CsPKS1_CAND010	CsPKS1	CsPKS1_REG04	180	0.5796	0.9009
6	5	BRD_CsPKS1_CAND013	CsPKS1	CsPKS1_REG03	180	0.5796	0.8335
7	6	BRD_CsPKS1_CAND006	CsPKS1	CsPKS1_REG01	250	0.5796	0.8772
8	8	BRD_CsSp1_CAND008	CsSp1	CsSp1_REG01	180	0.55	0.7524